

ResponseRally Software Requirements Specification (SRS)

Document Information

Document Title: ResponseRally Software Requirements Specification

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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for ResponseRally, an AI benchmarking and research interface that allows users to compare responses from multiple AI providers side-by-side with real-time streaming and performance metrics.

1.2 Document Conventions

- RFC 2119 keywords (MUST, SHOULD, MAY) are used to indicate requirement levels
- Requirements are numbered using hierarchical format (e.g., 2.1.1)
- Priority levels: P0 (Critical), P1 (High), P2 (Medium), P3 (Low)

1.3 Intended Audience

- Software architects and developers
- Product managers and stakeholders
- Quality assurance engineers
- Technical project managers

1.4 Product Scope

ResponseRally is a web-based application that provides a comprehensive platform for comparing AI model responses with real-time streaming, performance metrics, and export capabilities. The system consists of a React frontend and Node.js/Express backend with WebSocket support.

1.5 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- W3C Web Content Accessibility Guidelines (WCAG) 2.1
- OWASP Top 10 Security Guidelines

2. Overall Description

2.1 Product Perspective

ResponseRally is a standalone web application that serves as an AI response comparison platform. It integrates with multiple AI providers and provides real-time streaming capabilities, performance metrics, and export functionality.

2.2 Product Functions

- AI response comparison from multiple providers
- Real-time streaming display with progress indicators
- Performance metrics tracking and display
- Session management and conversation history
- Export functionality (PDF, JSON, CSV)
- Theme management (dark/light)

2.3 User Classes and Characteristics

User Class	Characteristics	Goals
AI Researchers	Technical expertise, frequent usage	Compare model responses, analyze performance
Developers	Technical background, integration needs	Evaluate AI capabilities for projects
Students	Learning-oriented, limited budgets	Understand AI model differences
Enterprise Users	Business-focused, security requirements	Assess AI solutions for deployment

2.4 Operating Environment

- Frontend: Modern browsers supporting ES2020+ (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)
- Backend: Node.js 18+ on Linux/Windows/macOS
- Deployment: Cloud platforms (AWS, Azure, GCP) or self-hosted
- Network: Internet connectivity required for AI provider APIs

2.5 Design and Implementation Constraints

- Must support real-time streaming with WebSocket fallback
- Client-side session state management with backend sync
- Responsive design for desktop and mobile
- Accessibility compliance (WCAG 2.1 AA)
- Security compliance with OWASP standards

2.6 Assumptions and Dependencies

- AI provider APIs remain available and stable
- Users have internet connectivity
- Browser supports modern JavaScript features
- WebSocket connections are available (fallback to polling if not)

3. External Interface Requirements

3.1 User Interfaces

3.1.1 Web Interface Requirements

- Layout: Responsive grid-based layout with header, main content, and footer
- Navigation: Intuitive navigation with clear section organization
- Theme: Support for dark/light themes with system preference detection
- Accessibility: Keyboard navigation, screen reader support, high contrast options

3.1.2 Component Requirements

- Header Component: Logo, theme toggle, export button, flow icon, session controls
- Prompt Input Component: Multi-line text input, provider toggles, submit button
- Response Grid Component: Horizontal scrolling grid for AI responses
- Response Card Component: Individual response display with metrics and actions
- Metrics Panel Component: Performance metrics display
- Conversation History Component: Scrollable history display

3.2 Hardware Interfaces

None specified. Application runs in web browsers on standard computing hardware.

3.3 Software Interfaces

3.3.1 Backend API Endpoints

Endpoint	Method	Purpose	Authentication
/api/v1/session	POST	Create new session	None
/api/v1/session/{id}	GET	Get session data	None
/api/v1/session/{id}/reset	POST	Reset session	None
/api/v1/prompt	POST	Submit prompt to providers	None
/api/v1/session/{id}/select-response	POST	Select best response	None
/api/v1/session/{id}/toggle-provider	POST	Toggle provider	None
/api/v1/session/{id}/retry-provider	POST	Retry provider	None
/health	GET	Health check	None

3.3.2 AI Provider APIs

- OpenRouter API: For GPT-4, LLaMA, Gemini, Copilot, DeepSeek via OpenRouter
- Mistral API: Direct integration with Mistral AI API
- WebSocket API: Real-time updates for streaming responses

3.4 Communications Interfaces

- HTTP/HTTPS: REST API communication between frontend and backend

- WebSocket: Real-time updates for streaming responses
- SSE (Server-Sent Events): Fallback for real-time updates

4. Functional Requirements

4.1 Session Management

4.1.1 Session Creation (REQ-001)

Priority: P0

Description: The system **MUST** create a new session when the application loads

Acceptance Criteria:

- New session ID is generated
- Default providers are enabled
- Empty conversation history is initialized
- Session metadata is set (created/updated timestamps)

4.1.2 Session Retrieval (REQ-002)

Priority: P1

Description: The system **MUST** retrieve session data by session ID

Acceptance Criteria:

- Session data is returned for valid session IDs
- 404 error is returned for invalid session IDs
- All session properties are included in response

4.1.3 Session Reset (REQ-003)

Priority: P1

Description: The system **MUST** reset a session to its initial state

Acceptance Criteria:

- Conversation history is cleared
- Current prompt is cleared
- Current responses are cleared
- Session metadata is updated

5. Non-Functional Requirements

5.1 Performance Requirements

5.1.1 Response Time (REQ-019)

Priority: P0

Requirement: The system MUST respond to user actions within 2 seconds

5.2 Security Requirements

5.2.1 Data Protection (REQ-022)

Priority: P0

Requirement: The system MUST NOT store user prompts or responses permanently

9. Appendices

Appendix A: Glossary

AI Provider: Service that generates AI responses (GPT-4, LLaMA, etc.)

Session: Individual user interaction context with conversation history

Streaming: Real-time delivery of response content as it's generated

Metrics: Quantitative measurements of response performance

Document Version History

v1.0 (Jan 16, 2026): Initial release